Please check the examination details below before entering your candidate information

Candidate surname	Other names

Pearson Edexcel
Level 1/Level 2 GCSE (9–1)

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Hannah Kettle MATHS TUTORIALS



Half Past Paper – Week 22

Higher Tier

Total Marks

Q	Topic	Marks	Sparx Code
2a	Finding the highest common factor (HCF)	2	U529
2b	Finding the lowest common multiple (LCM)	2	U751
4a	Plotting graphs of quadratic functions	2	U989
4b	Plotting graphs of quadratic functions	2	U989
4c	Solving quadratic equations graphically	2	U601
6a	Finding unknown sides in right-angled triangles	2	U283
6b	Finding unknown angles in right-angled triangles	2	U545
8	Drawing cumulative frequency graphs	1	U182
10	Compound interest calculations, Growth and decay	4	U332,U988
12	Expanding triple brackets	3	U606
14	Alternate segment theorem, Mixed problems: Circle theorems	4	U130,U808
16a	Substituting into iterative formulae	3	U434
16b	Substituting into iterative formulae	1	U434
18a	Graphs of trigonometric functions	2	U450
18b	Transforming graphs	1	U455
18c	Transforming graphs	1	U455
20	Conditional probabilities from tree diagrams	4	U806
	Total	38	

Answer ALL questions.

Write your answers in the spaces provided.

You must write down all the stages in your working.

2 (*a*) Find the Highest Common Factor (HCF) of 60 and 84

.....

(2)

(*b*) Find the Lowest Common Multiple (LCM) of 24 and 40

.....

(2)

(Total for Question 2 is 4 marks)

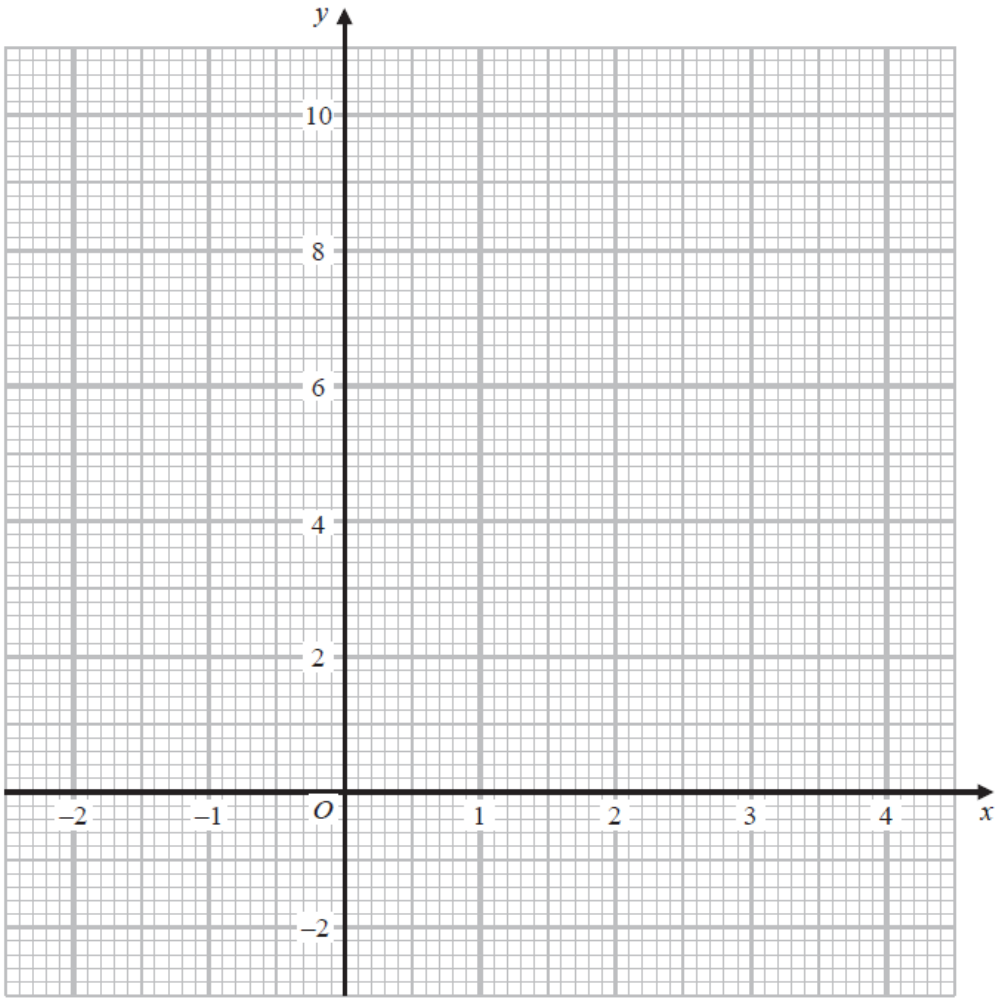
4 (*a*) Complete the table of values for $y = x^2 - 2x + 2$

x	–2	–1	0	1	2	3	4
y	10		2			5	

(2)

(*b*) On the grid, draw the graph of $y = x^2 - 2x + 2$ for values of x from –2 to 4

(2)



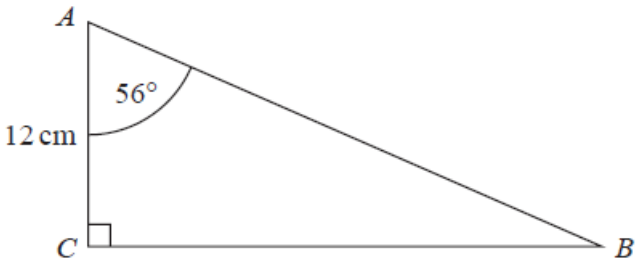
(*c*) Use your graph to find estimates of the solutions of the equation $x^2 - 2x + 2 = 4$

.....

(2)

(Total for Question 4 is 6 marks)

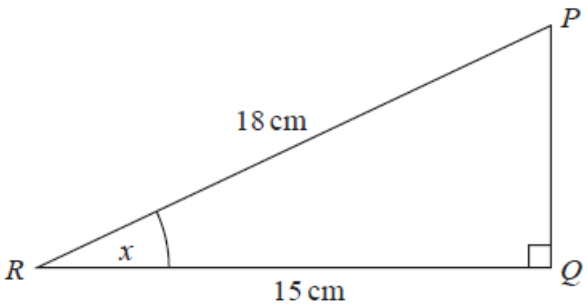
6 *ABC* is a right-angled triangle.



(*a*) Work out the length of *BC*.
Give your answer correct to 1 decimal place.

..... cm
(2)

PQR is a right-angled triangle.



(*b*) Work out the size of the angle marked *x*.
Give your answer correct to 1 decimal place.

.....°
(2)

(Total for Question 6 is 4 marks)

8 The grouped frequency table gives information about the time, in minutes, taken by 50 people to solve a puzzle.

Time (t minutes)	**Frequency**
$0 < t \leq 10$	5
$10 < t \leq 20$	8
$20 < t \leq 30$	12
$30 < t \leq 40$	15
$40 < t \leq 50$	7
$50 < t \leq 60$	3

Brian was asked to draw a cumulative frequency table for this information.

This is the table that Brian drew.

Time (t minutes)	**Cumulative frequency**
$0 < t \leq 10$	5
$10 < t \leq 20$	13
$20 < t \leq 30$	25
$30 < t \leq 40$	40
$40 < t \leq 50$	47
$50 < t \leq 60$	50

Write down **one** thing that is wrong with this cumulative frequency table.

.....

.....

.....

(Total for Question 8 is 1 mark)

10 Louise invests £x in Better Investments for 3 years.
Sadiq invests £x in County Bank for 3 years.

Better Investments
Compound Interest
2.5% per annum

County Bank
Compound Interest
2% per annum for the first two years 3.5% per annum for each extra year

At the end of the 3 years, the value of Louise's investment is £344 605

Work out the value of Sadiq's investment at the end of the 3 years.

£.....

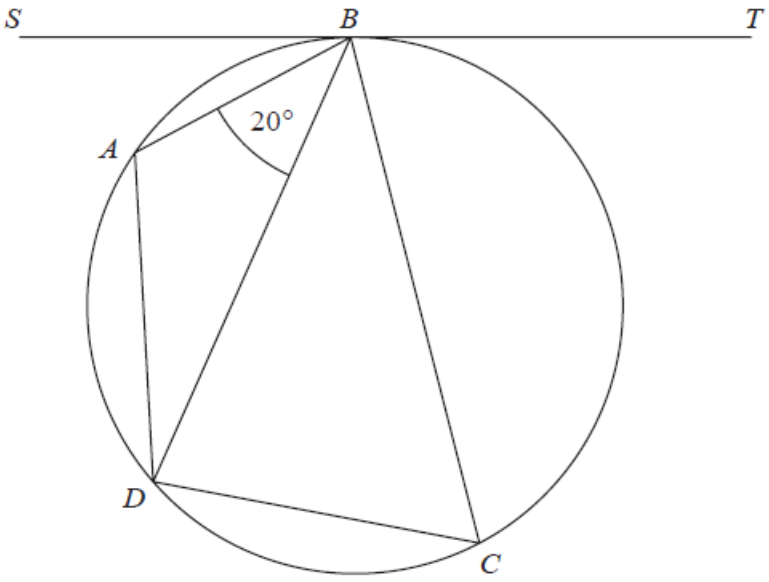
(Total for Question 10 is 4 marks)

12 Expand and simplify $(x - 2)(3x + 2)(2x + 3)$

.....

(Total for Question 12 is 3 marks)

14



A, *B*, *C* and *D* are four points on a circle.
SBT is a tangent to the circle.
Angle *ABD* = 20°

the size of angle *BAD* : the size of angle *BCD* = 3 : 1

Find the size of angle *SBA*.
Give a reason for each stage of your working.

.....°

(Total for Question 14 is 4 marks)

16 (*a*) Use the iteration formula $x_{n+1} = \sqrt[3]{10 - 2x_n}$ to find the values of x_1, x_2 and x_3
Start with $x_0 = 2$

x_1 = ...

x_2 = ...

x_3 = ...

(3)

The values of x_1, x_2 and x_3 found in part (*a*) are estimates of the solution of an equation of the form $x^3 + ax + b = 0$ where a and b are integers.

(*b*) Find the value of a and the value of b.

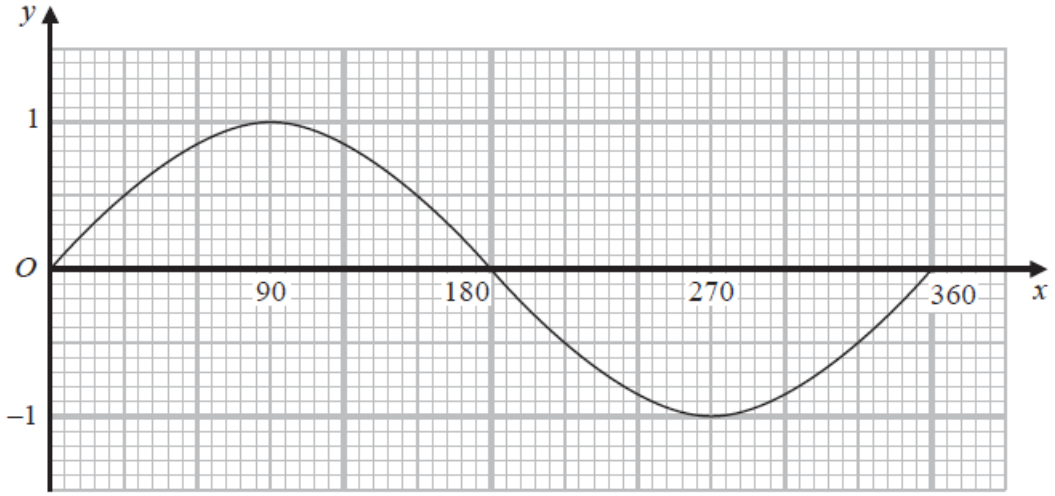
a = ...

b = ...

(1)

(Total for Question 16 is 4 marks)

18 Here is a graph of $y = \sin x°$ for $0 \leq x \leq 360$



(*a*) Using this graph, find estimates of all **four** solutions of

$$\sin x° = 0.6 \text{ for } 0 \leq x \leq 720$$

.....

(2)

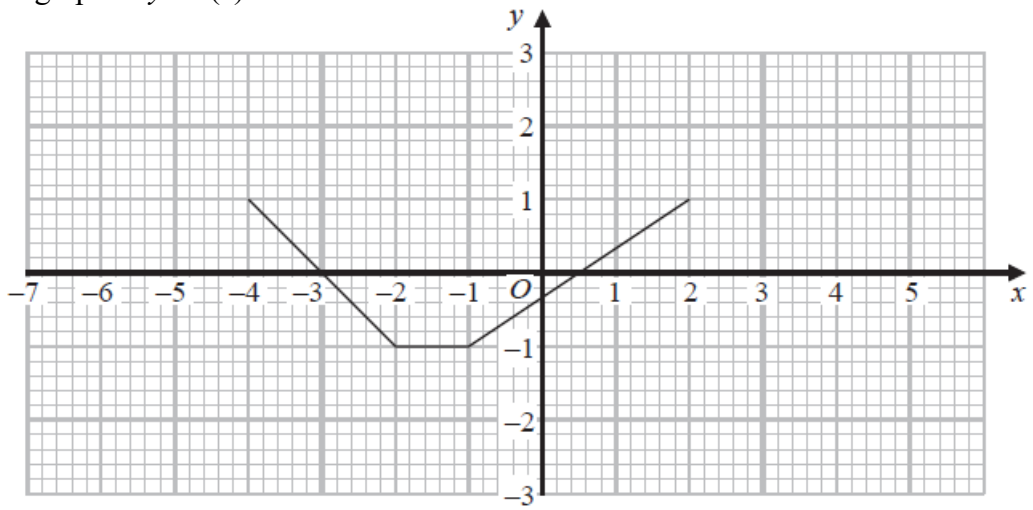
The graph of $y = \sin x°$ is reflected in the x-axis.

(*b*) Write down an equation of the reflected graph.

.....

(1)

Here is a graph of $y = f(x)$



(*c*) On the grid, draw the graph of $y = f(x - 2)$

(1)

(Total for Question 18 is 4 marks)

20 In a village,

if it rains on one day, the probability that it will rain on the next day is 0.8
if it does **not** rain on one day, the probability that it will rain on the next day is 0.6

A weather forecaster says,

"There is a 70% chance that it will rain in the village on Monday."

Work out an estimate for the probability that it will rain in the village on Wednesday.
You must show all your working.

.....

(Total for Question 20 is 4 marks)

TOTAL FOR PAPER IS 38 MARKS